

T1486 - Data Encrypted for Impact Data Encrypted with GPG4Win

1. Executive Summary

Ngày 28/07/2026, Atomic Red Team T1486 Test #8 được thực hiện trên máy Windows 10 victim nhằm mô phỏng hành vi mã hóa dữ liệu bằng GPG4Win.

LimaCharlie Timeline ghi nhận chuỗi tiến trình:

```
powershell.exe
└─ cmd.exe
    └─ gpg.exe
```

Command-line telemetry cho thấy gpg.exe được gọi với chế độ mã hóa đối xứng, chế độ không tương tác và một passphrase được truyền trực tiếp trên command line. Tiến trình nhắm tới file:

```
C:\Users\longday\AppData\Local\Temp\test.txt
```

Kết luận: True Positive - Authorized Simulation.

2. Incident Classification

Hạng mục	Kết quả
Case Name	Data Encryption Activity Using GPG4Win
Classification	True Positive - Authorized Simulation
MITRE Tactic	Impact
MITRE Technique	T1486 - Data Encrypted for Impact
Atomic Test	T1486 Test #8 - Data Encrypted with GPG4Win
Affected Host	DESKTOP-JI0TPU0
Executing User	DESKTOP-JI0TPU0\longday
Source IP	Not Evidenced
Execution Date	28/07/2026
Core Activity Time	21:33:19 - 21:33:24 ICT
Analyst Timeline Source	LimaCharlie Timeline
Detection Severity	High
Observed System Impact	Limited and Reversible
Business Impact	None
Containment	Not Required
Prevention Result	Not Observed
Case Status	Close after cleanup validation

3. Test Objective and Scope

Phạm vi chỉ bao gồm một file thử nghiệm trong thư mục TEMP của người dùng:
C:\Users\longday\AppData\Local\Temp\test.txt

4. Test Execution and Detection Summary

-c xác nhận yêu cầu mã hóa đối xứng. Không nên ghi AES-256 nếu không có output của GnuPG hoặc metadata artifact xác nhận thuật toán.

```
*****BEGIN TEST*****
Technique: Data Encrypted for Impact T1486
Atomic Test Name: Data Encrypted with GPG4Win
Atomic Test Number: 8
Atomic Test GUID: 4541e2c2-33c8-44b1-be79-9161440F1718
Description: Gpg4win is a Windows tool (also called Kleopatra which is the preferred certificate manager) that uses email and file encryption packages for symmetric encryption. It is used by attackers to encrypt disks. User will need to add pass phrase to encrypt file as automation is not allowed under newer versions.

Attack Commands:
Executor: powershell
ElevationRequired: False
Command:
Set-Content -Path "%(FileToEncryptLocation)" -Value "populating this file with some text" # Create the test.txt file again in case prereqs failed
cmd /c "C:\Program Files (x86)\GnuPG\bin\gpg.exe" --passphrase "SomeParaphraseBlah" --batch --yes -c "%(FileToEncryptLocation)"
Command (with inputs):
Set-Content -Path "%env:temp\test.txt" -Value "populating this file with some text" # Create the test.txt file again in case prereqs failed
cmd /c "C:\Program Files (x86)\GnuPG\bin\gpg.exe" --passphrase "SomeParaphraseBlah" --batch --yes -c "%env:temp\test.txt"

Cleanup Commands:
Command:
Remove-Item -Path "%(FileToEncryptLocation)" -Force -ErrorAction SilentlyContinue
Remove-Item -Path "%(FileToEncryptLocation).gpg" -Force -ErrorAction SilentlyContinue
Command (with inputs):
Remove-Item -Path "%env:temp\test.txt" -Force -ErrorAction SilentlyContinue
Remove-Item -Path "%env:temp\test.txt.gpg" -Force -ErrorAction SilentlyContinue

Dependencies:
Description: GPG must exist at (C:\Program Files (x86)\GnuPG\bin\gpg.exe). If -GetPrereqs fails, try to install GPG4Win manually at 'https://www.gpg4win.org/download.html'. Once done, run -CheckPrereqs to confirm that it works.
Check Prereq Command:
If (test-path "%(GPG_ExecLocation)") {exit 0} else {exit 1}
Check Prereq Command (with inputs):
If (test-path "C:\Program Files (x86)\GnuPG\bin\gpg.exe") {exit 0} else {exit 1}
Get Prereq Command:
Set-Content -Path "%(FileToEncryptLocation)" -Value "populating this file with some text" # Create the test.txt file
New-Item -Type Directory "PathToAtomicFolder\..\ExternalPayloads\" -ErrorAction Ignore -Force | Out-Null
Invoke-WebRequest "https://files.gpg4win.org/gpg4win-4.1.0.exe" -outfile "PathToAtomicFolder\..\ExternalPayloads\gpginstall.exe"
cmd /c "PathToAtomicFolder\..\ExternalPayloads\gpginstall.exe" /S
Set Prereq Command (with inputs):
Set-Content -Path "%env:temp\test.txt" -Value "populating this file with some text" # Create the test.txt file
New-Item -Type Directory "C:\AtomicRedTeam\atomic\..\ExternalPayloads\" -ErrorAction Ignore -Force | Out-Null
Invoke-WebRequest "https://files.gpg4win.org/gpg4win-4.1.0.exe" -outfile "C:\AtomicRedTeam\atomic\..\ExternalPayloads\gpginstall.exe"
cmd /c "C:\AtomicRedTeam\atomic\..\ExternalPayloads\gpginstall.exe" /S
[!!!!!!]END TEST!!!!!!
```

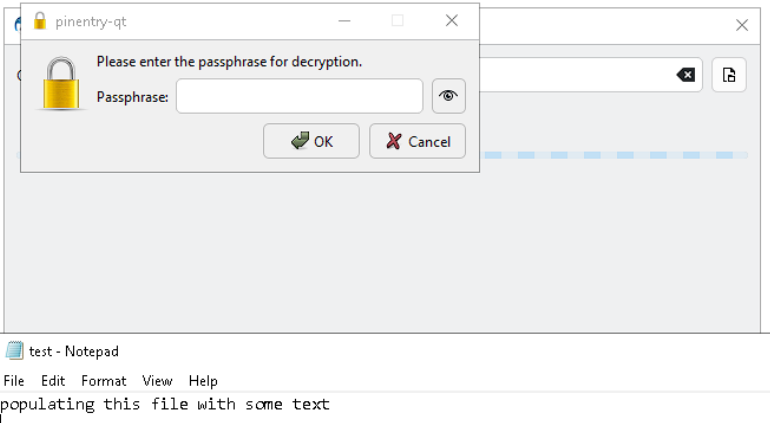
```
21:33:20 Suspicious Execution of Hostname → desktop-j10tpu0 {"event":{"COMMAND_LINE":"","C:\Windows\system32\HOSTNAME.EXE","FILE_IS_SIGNED":1,"FILE_PATH":"C:\Windows\system32\HOSTNAME.EXE","HASH":
21:33:20 Suspicious Execution of Hostname → desktop-j10tpu0 {"event":{"COMMAND_LINE":"","C:\Windows\system32\HOSTNAME.EXE","FILE_IS_SIGNED":1,"FILE_PATH":"C:\Windows\system32\HOSTNAME.EXE","HASH":
21:33:20 Non Interactive PowerShell Process Spammed → desktop-j10tpu0 {"event":{"BASE_ADDRESS":"140694585409536","COMMAND_LINE":"","powershell.exe" & {Set-Content -Path ""$env:temp\test.txt"" -Val
21:33:19 Local Accounts Discovery → desktop-j10tpu0 {"event":{"COMMAND_LINE":"","C:\Windows\system32\whoami.exe","FILE_IS_SIGNED":1,"FILE_PATH":"C:\Windows\system32\whoami.exe","HASH":"1d4902a04d
21:33:19 Suspicious Execution of Hostname → desktop-j10tpu0 {"event":{"COMMAND_LINE":"","C:\Windows\system32\HOSTNAME.EXE","FILE_IS_SIGNED":1,"FILE_PATH":"C:\Windows\system32\HOSTNAME.EXE","HASH":
21:33:05 Suspicious Execution of Hostname → desktop-j10tpu0 {"event":{"COMMAND_LINE":"","C:\Windows\system32\HOSTNAME.EXE","FILE_IS_SIGNED":1,"FILE_PATH":"C:\Windows\system32\HOSTNAME.EXE","HASH":
21:33:05 Suspicious Execution of Hostname → desktop-j10tpu0 {"event":{"COMMAND_LINE":"","C:\Windows\system32\HOSTNAME.EXE","FILE_IS_SIGNED":1,"FILE_PATH":"C:\Windows\system32\HOSTNAME.EXE","HASH":
21:33:05 Non Interactive PowerShell Process Spammed → desktop-j10tpu0 {"event":{"BASE_ADDRESS":"140694585409536","COMMAND_LINE":"","powershell.exe" & {if (test-path "C:\Program Files (x86)\GnuPG\bin\gr
21:33:04 Local Accounts Discovery → desktop-j10tpu0 {"event":{"COMMAND_LINE":"","C:\Windows\system32\whoami.exe","FILE_IS_SIGNED":1,"FILE_PATH":"C:\Windows\system32\whoami.exe","HASH":"1d4902a04d
21:33:04 Suspicious Execution of Hostname → desktop-j10tpu0 {"event":{"COMMAND_LINE":"","C:\Windows\system32\HOSTNAME.EXE","FILE_IS_SIGNED":1,"FILE_PATH":"C:\Windows\system32\HOSTNAME.EXE","HASH":
```

```
21:33:04 ✓ NEW_PROCESS Process (PID): HOSTNAME.EXE (5972) Path: C:\Windows\system32\HOSTNAME.EXE Command: "C:\Windows\system32\HOSTNAME.EXE" User: {"COMMAND_LINE":"","C:\Windows\system32\HOSTNAME.EXE","FILE_IS
21:33:04 ✓ NEW_PROCESS Process (PID): whoami.exe (5972) Path: C:\Windows\system32\whoami.exe Command: "C:\Windows\system32\whoami.exe" User: {"COMMAND_LINE":"","C:\Windows\system32\whoami.exe","FILE_IS_SIGNED":
21:33:05 ✓ NEW_PROCESS Process (PID): powershell.exe (6124) Path: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe Command: "powershell.exe" & {if (test-path "C:\Program Files (x86)\GnuPG\bin\gpg.exe") {exit 1
21:33:05 ✓ NEW_PROCESS Process (PID): Conhost.exe (7755) Path: C:\Windows\System32\Conhost.exe Command: "C:\Windows\system32\conhost.exe 0xffffffff ForceV1 User: DESKTOP-J10TPU0\longday {"BASE_ADDRESS":"14069724
21:33:05 ✓ NEW_PROCESS Process (PID): HOSTNAME.EXE (3464) Path: C:\Windows\system32\HOSTNAME.EXE Command: "C:\Windows\system32\HOSTNAME.EXE" User: {"COMMAND_LINE":"","C:\Windows\system32\HOSTNAME.EXE","FILE_IS
21:33:05 ✓ NEW_PROCESS Process (PID): HOSTNAME.EXE (5592) Path: C:\Windows\system32\HOSTNAME.EXE Command: "C:\Windows\system32\HOSTNAME.EXE" User: {"COMMAND_LINE":"","C:\Windows\system32\HOSTNAME.EXE","FILE_IS
21:33:05 TERMINATE_PROCESS PID: 7976 {"PARENT_PROCESS_ID":5656,"PROCESS_ID":7976}
21:33:05 TERMINATE_PROCESS PID: 7755 {"PARENT_PROCESS_ID":6124,"PROCESS_ID":7755}
21:33:05 TERMINATE_PROCESS PID: 5972 {"PARENT_PROCESS_ID":5656,"PROCESS_ID":5972}
21:33:05 TERMINATE_PROCESS PID: 6124 {"PARENT_PROCESS_ID":5656,"PROCESS_ID":6124}

21:33:19 ✓ NEW_PROCESS Process (PID): HOSTNAME.EXE (2776) Path: C:\Windows\system32\HOSTNAME.EXE Command: "C:\Windows\system32\HOSTNAME.EXE" User: {"COMMAND_LINE":"","C:\Windows\system32\HOSTNAME.EXE","FILE_IS
21:33:19 ✓ NEW_PROCESS Process (PID): whoami.exe (1516) Path: C:\Windows\system32\whoami.exe Command: "C:\Windows\system32\whoami.exe" User: {"COMMAND_LINE":"","C:\Windows\system32\whoami.exe","FILE_IS_SIGNED":
21:33:20 ✓ NEW_PROCESS Process (PID): powershell.exe (7236) Path: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe Command: "powershell.exe" & {Set-Content -Path ""$env:temp\test.txt"" -Value ""populating
21:33:20 ✓ NEW_PROCESS Process (PID): Conhost.exe (7012) Path: C:\Windows\System32\Conhost.exe Command: "C:\Windows\system32\conhost.exe 0xffffffff ForceV1 User: DESKTOP-J10TPU0\longday {"BASE_ADDRESS":"14069724
21:33:20 NEW_DOCUMENT Path: C:\Users\longday\AppData\Local\Temp\__PSScriptPolicyTest_kttz44k.szm.ps1 PID: 7236 {"ERROR":2,"FILE_PATH":"C:\Users\longday\AppData\Local\Temp\__PSScriptPolicyTest_kttz44k.szm.ps1"
21:33:20 NEW_DOCUMENT Path: C:\Users\longday\AppData\Local\Temp\__PSScriptPolicyTest_kttz44k.szm.ps1 PID: 7236 {"ERROR":2,"FILE_PATH":"C:\Users\longday\AppData\Local\Temp\__PSScriptPolicyTest_kttz44k.szm.ps1"
21:33:20 ✓ NEW_PROCESS Process (PID): HOSTNAME.EXE (4360) Path: C:\Windows\system32\HOSTNAME.EXE Command: "C:\Windows\system32\HOSTNAME.EXE" User: DESKTOP-J10TPU0\longday {"COMMAND_LINE":"","C:\Windows\system32\H
21:33:20 ✓ NEW_PROCESS Process (PID): HOSTNAME.EXE (4795) Path: C:\Windows\system32\HOSTNAME.EXE Command: "C:\Windows\system32\HOSTNAME.EXE" User: DESKTOP-J10TPU0\longday {"COMMAND_LINE":"","C:\Windows\system32\H
21:33:20 ✓ NEW_PROCESS Process (PID): cmd.exe (2032) Path: C:\Windows\system32\cmd.exe Command: "C:\Windows\system32\cmd.exe" /c "C:\Program Files (x86)\GnuPG\bin\gpg.exe" --passphrase "SomeParaphraseBlah" --batch --
21:33:20 ✓ NEW_PROCESS Process (PID): gpg.exe (7472) Path: C:\Program Files (x86)\GnuPG\bin\gpg.exe Command: "C:\Program Files (x86)\GnuPG\bin\gpg.exe" --passphrase "SomeParaphraseBlah" --batch --yes -c "C:\Users\lon
21:33:21 ✓ CODE_IDENTITY Hash: 7711b4c4bfed5809745806915ce50a48b5b2ba0b69a5f4c1e67d445e5e Path: C:\Program Files (x86)\GnuPG\bin\gpg.exe {"ACCESS_TIME":1785246871899,"ATTRIBUTES":0,"CREATION_TIME":1771929760000,"EI
21:33:21 TERMINATE_PROCESS PID: 2776 {"PARENT_PROCESS_ID":5656,"PROCESS_ID":2776}
21:33:21 TERMINATE_PROCESS PID: 1516 {"PARENT_PROCESS_ID":5656,"PROCESS_ID":1516}
21:33:21 TERMINATE_PROCESS PID: 4360 {"PARENT_PROCESS_ID":7236,"PROCESS_ID":4360}
21:33:21 TERMINATE_PROCESS PID: 7472 {"PARENT_PROCESS_ID":2032,"PROCESS_ID":7472}
21:33:21 TERMINATE_PROCESS PID: 7012 {"PARENT_PROCESS_ID":7236,"PROCESS_ID":7012}
21:33:21 TERMINATE_PROCESS PID: 7236 {"PARENT_PROCESS_ID":5656,"PROCESS_ID":7236}
21:33:21 TERMINATE_PROCESS PID: 4795 {"PARENT_PROCESS_ID":7236,"PROCESS_ID":4795}
21:33:21 TERMINATE_PROCESS PID: 2032 {"PARENT_PROCESS_ID":7236,"PROCESS_ID":2032}

21:33:24 ✓ NETWORK_CONNECTIONS Connections: 1 Process (PID): splunkd.exe (5100) {"COMMAND_LINE":"","C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe" service","CREATION_TIME":1785246873074,"FILE_IS_SIGNED":1,"FIL
21:33:24 ✓ NETWORK_CONNECTIONS Connections: 1 Process (PID): gpg.exe (7472) {"COMMAND_LINE":"","C:\Program Files (x86)\GnuPG\bin\gpg.exe" --passphrase "SomeParaphraseBlah" --batch --yes -c "C:\Users\longday\AppData\Li
21:33:24 ✓ NETWORK_CONNECTIONS Connections: 3 Process (PID): gpg-agent.exe (7300) {"COMMAND_LINE":"","C:\Program Files (x86)\GnuPG\bin\gpg-agent.exe" --homedir C:\Users\longday\AppData\Roaming\gnupg --daemon","FILE_IS_SIGNI
21:33:24 ✓ NETWORK_CONNECTIONS Connections: 2 Process (PID): keyboard.exe (5040) {"COMMAND_LINE":"","C:\Program Files\GnuPG\bin\keyboard.exe" --homedir C:\Users\longday\AppData\Roaming\gnupg --daemon","FILE_IS_SIGNED":
21:33:14 TERMINATE_PROCESS PID: 6164 {"PARENT_PROCESS_ID":788,"PROCESS_ID":6164}
21:33:14 TERMINATE_PROCESS PID: 7340 {"PARENT_PROCESS_ID":788,"PROCESS_ID":7340}
21:33:12 TERMINATE_PROCESS PID: 444 {"PARENT_PROCESS_ID":656,"PROCESS_ID":444}
21:33:15 ✓ NETWORK_CONNECTIONS Connections: 2 Process (PID): splunkd.exe (5100) {"COMMAND_LINE":"","C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe" service","CREATION_TIME":1785246873074,"FILE_IS_SIGNED":1,"FIL
21:33:15 ✓ NETWORK_CONNECTIONS Connections: 2 Process (PID): splunkd.exe (5100) {"COMMAND_LINE":"","C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe" service","CREATION_TIME":1785246873074,"FILE_IS_SIGNED":1,"FIL
```

notificationimages	6/29/2026 4:06 PM	File folder
scoped_dir1656_128114903	7/16/2026 1:11 PM	File folder
scoped_dir2172_146846909	7/8/2026 5:31 PM	File folder
scoped_dir4496_588699524	7/14/2026 12:35 AM	File folder
scoped_dir5276_783411261	7/7/2026 12:47 AM	File folder
scoped_dir5500_1156060720	7/15/2026 3:59 PM	File folder
scoped_dir6208_1727989545	7/13/2026 12:13 PM	File folder
scoped_dir6352_621991687	7/11/2026 10:52 AM	File folder
scoped_dir8464_1656267582	6/29/2026 4:06 PM	File folder
scoped_dir10436_196874712	7/13/2026 12:13 AM	File folder
vmware-longday	6/15/2026 9:56 PM	File folder
WinGet	7/9/2026 1:24 AM	File folder
WinRAR.exe.WebView2	7/9/2026 1:40 AM	File folder
WinSAT	4/22/2026 9:41 AM	File folder
zui_pastes_zzV9EF	4/3/2026 8:42 PM	File folder
3a6be0cc-b7a9-4f2b-809f-0c3359f8fd90t...	7/28/2026 9:23 PM	TMP File
2132adac-bf46-4c20-aacf-7c70dad18a5a...	7/28/2026 9:23 PM	TMP File
cc40cb78-05d4-4693-baa6-1fc58b83bd4e...	7/28/2026 9:24 PM	TMP File
d6d393bc-f19a-4f37-8477-27f4e1d25a64t...	7/28/2026 9:23 PM	TMP File
Invoke-AtomicTest-ExecutionLog.csv	7/28/2026 9:33 PM	CSV File
test	7/28/2026 9:33 PM	Text Document
test.txt	7/28/2026 9:33 PM	OpenPGP Binary F...
tmp45CE.tmp	7/27/2026 3:30 PM	TMP File



5. TimeLine Analyst

Thời gian	Process	Command / Action
21:33:04	HOSTNAME.EXE	HOSTNAME.EXE
21:33:04	whoami.exe	whoami.exe
21:33:05	powershell.exe	"powershell.exe" & {if (test-path 'C:\Program Files (x86)\GnuPG\bin\gpg.exe'){exit 0} else {exit 1}}
		Kiểm tra xem hệ thống có GPG4Win 32 bit chưa
21:33:19	whoami.exe	whoami.exe
21:33:20	powershell.exe	"powershell.exe" & {Set-Content -Path \"\$env:temp\test.txt\" -Value \"\$env:temp\test.txt\" # Create the test.txt file again in case prereqs failed <i>Tạo file test.txt trong thư mục TEMP kèm nội dung</i> cmd /c \"\$env:temp\test.txt\" <i>Dùng cmd mở gpg.exe</i> --passphrase 'SomeParaphraseBlah' --batch --yes -c \"\$env:temp\test.txt\" <i>--passphrase cung cấp passphrase trực tiếp qua command-line thay vì để gpg hỏi tương tác qua prompt/pinentry dialog.</i> <i>--batch tắt hoàn toàn chế độ tương tác — gpg sẽ không hỏi bất kỳ câu gì (confirm, warning...), chạy thẳng theo tham số đã cho.</i> <i>--yes tự động trả lời "yes" cho mọi câu hỏi xác nhận overwrite — cụ thể ở đây là nếu file output test.txt.gpg đã tồn tại từ lần chạy trước, gpg sẽ ghi đè luôn mà không hỏi.</i> <i>-c viết tắt của mã hóa symetric. Thuật toán mặc định của GnuPG hiện tại là AES-256.</i>
21:33:20	powershell.exe	PSScriptPolicyTest_kttzd4ak.szm.ps1
21:33:20	cmd.exe	/c \"\$env:temp\test.txt\" --passphrase 'SomeParaphraseBlah' --batch --yes -c \"\$env:temp\test.txt\"
21:33:20	gpg.exe	--passphrase 'SomeParaphraseBlah' --batch --yes -c \"\$env:temp\test.txt\"
21:33:21	gpg.exe	SHA256 hash của gpg.exe
21:33:24	gpg.exe	Connections: 1
21:33:24	gpg-agent.exe	--homedir ... \gnupg --daemon <i>Là bên xử lý yêu cầu và giao tiếp với gpg.exe qua local IPC vì gpg.exe không tự quản lý key/passphrase trực tiếp nữa mà giao việc đó cho một daemon nền chạy riêng là gpg-agent</i>
21:33:24	keyboxd.exe	--homedir ... \gnupg --daemon <i>keyboxd.exe là nơi lưu trữ public key thay cho định dạng pubring.gpg cũ. Nó tách riêng việc quản lý key storage ra khỏi agent để modular hóa.</i>
21:35:29	gpg-agent.exe	Connections: 2
21:35:29	keyboxd.exe	Connections: 2

